

Robotics Camp 2026 Capstone

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Overview

With the conclusion of my work in Robotics Camp during the summer of 2026, I have decided to create a pong game as a capstone of my learning experience.

In this presentation, I will provide insights into both the planning and execution steps of this project.

More detailed information can be found in the project docs.

Planning

Concept - Make a fully functional ping-pong game

Analysis - 2 OLED screens, IR sensor and controller, joystick, Pico 2W and Breadboard, Jumper Wires. Code must have functions for drawing, detecting IR, moving paddles, ball physics, storing scores, choosing modes.

Requirements - Must show the proper layout, enable paddle movement, make the ball move in a realistic manner, detect collisions between both paddles and walls, award points where needed, and detect the winner

Design - Circuit schematics, build layout

Execution

Build - Construct circuit and code, refine if test fails or if there is room for improvement

Test - Run program and fix errors as needed, also change parts of the code to better fit the prompt

Present - Share final working result

Docs

Demonstration

THANK YOU!